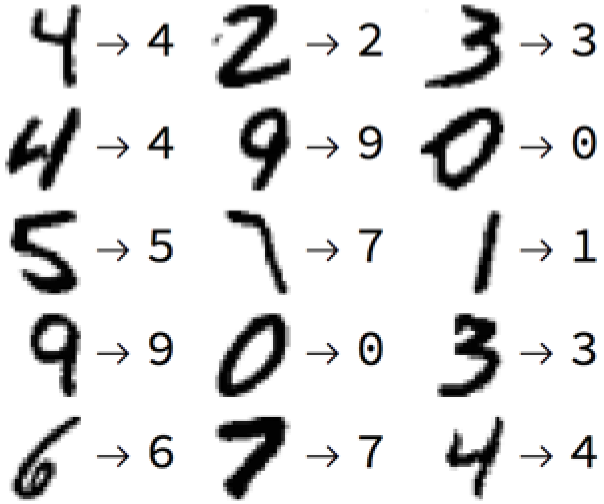
Clustering

Types of Learning: Supervised

"It's a field of study that gives the ability to the computer to self-learn without being explicitly programmed"

- Input: $x \in X$
- Output (label): $y \in Y$
- Target: $f\colon X \to Y$
- Data: $(x_1, y_1), (x_2, y_2), (x_3, y_3), \ldots, (x_N, y_N)$

- Hypothesis: $g\colon X \to Y$



Active learning:

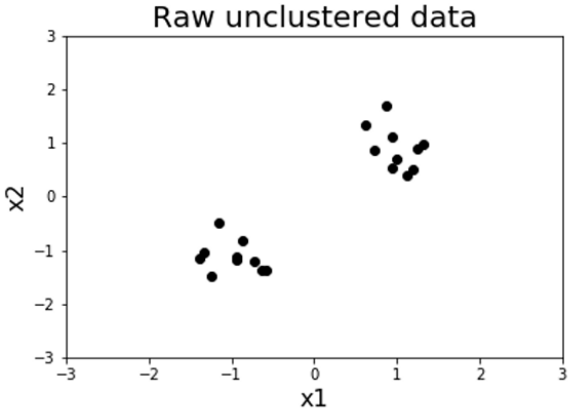
– There is $O\colon X \to Y$ – an oracle. We want to get g with a minimum number of request to O

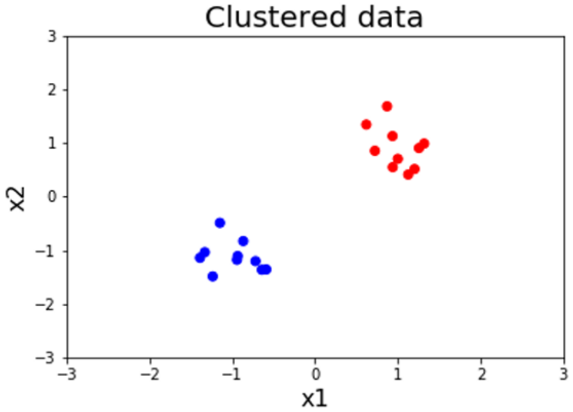
Online learning (but not only for supervised learning):

– The dataset is given to the algorithm by one example at time (example: recommendation system)

Types of Learning: Unsupervised

- Input: $x \in X$
- Data: $\mathrm{x}_1, x_2, x_3, \ldots, x_N$





We have sample points, but no labels!
No classes, no y-values, nothing to predict

Goal: Discover structure in the data

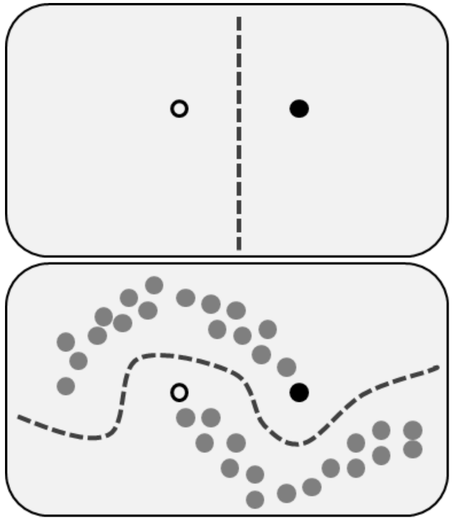
Examples:

- Clustering – partition data into groups of similar/nearby points
- Dimensionality reduction – data often lies near a low-dimensional subspace (or manifold) in feature space; matrices have low-rank approximations.
- Density estimation: fit a continuous distribution to discrete data.

Types of Learning: Semi-Supervised

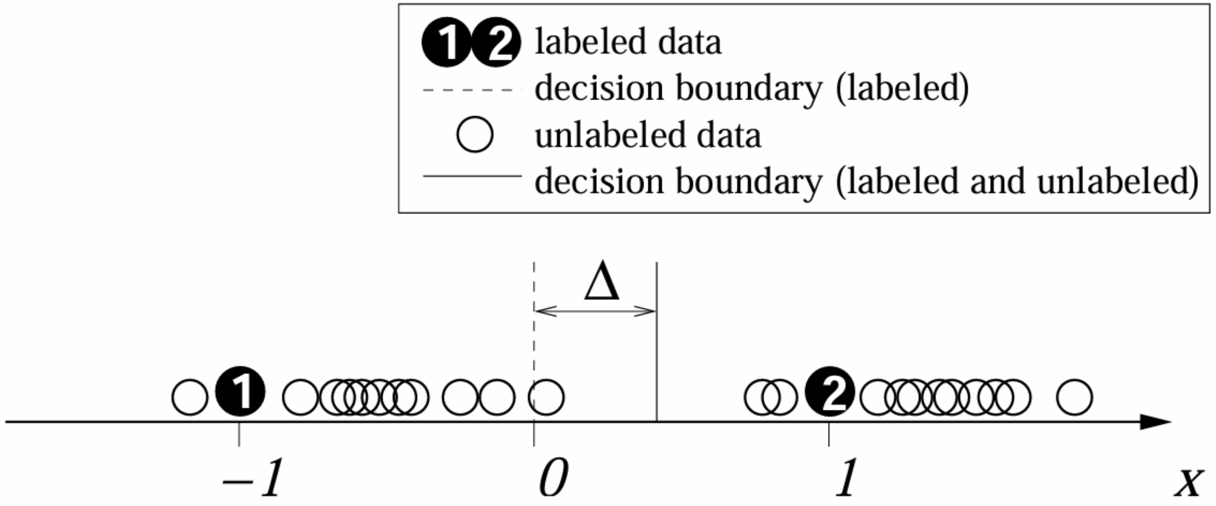
- Input: $x \in X$
- Output (label): $y \in Y$
- Target: $f: X \to Y$
- Labeled data: $(x_1, y_1), (x_2, y_2), (x_3, y_3), \ldots, (x_N, y_N)$
- Unlabeled data: $\mathrm{x}_1, x_2, x_3, \ldots, x_N$

- Hypothesis: $g: X \to Y$
- Goal: find best hypothesis with using both type of the data



Example: Self-training algorithm
Assumption: one's own high confidence predictions are correct.

- Train f from D_l
- Predict on $x \in D_u$
- Add $(x, f(x))$ to labeled data
- Repeat



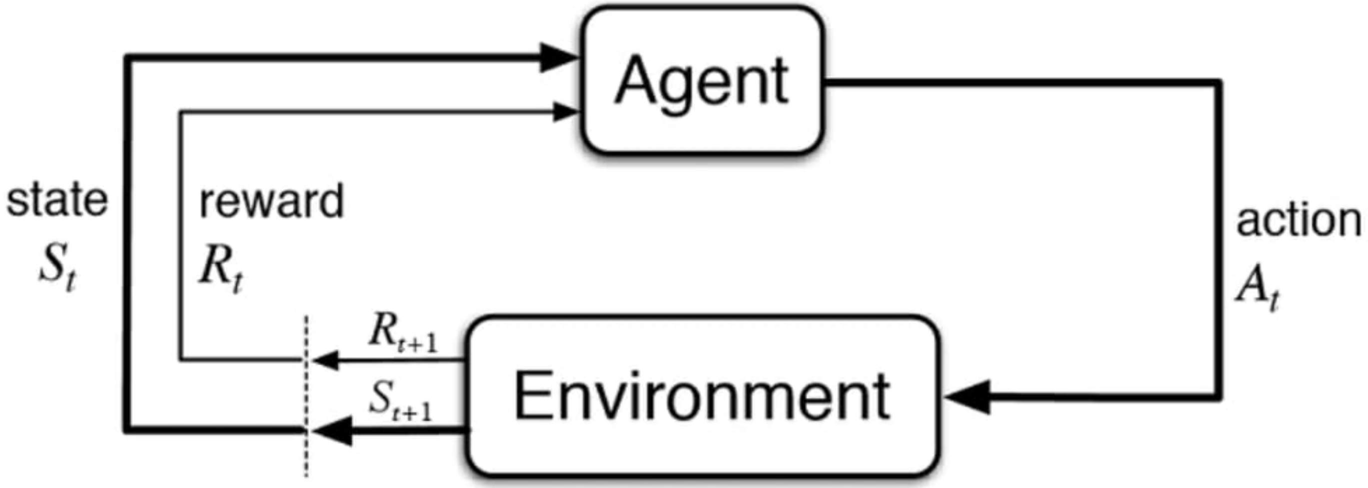
Types of Learning: Reinforcement

- Input: $s_t \in S$
- Output: $a_t \in A$
- Data: $(s_1, a_1, r_1, \ldots, s_T, a_T, r_T)$

Goal: learn $\pi_\theta : S \to A$ to maximize $\sum_t \gamma^t r_t$, $\gamma \in (0, 1]$

No supervision, only a reward signal!

Examples:
- Fly a helicopter
- Manage an investment portfolio
- Control a power station
- Make a robot walk
- Play video or board games

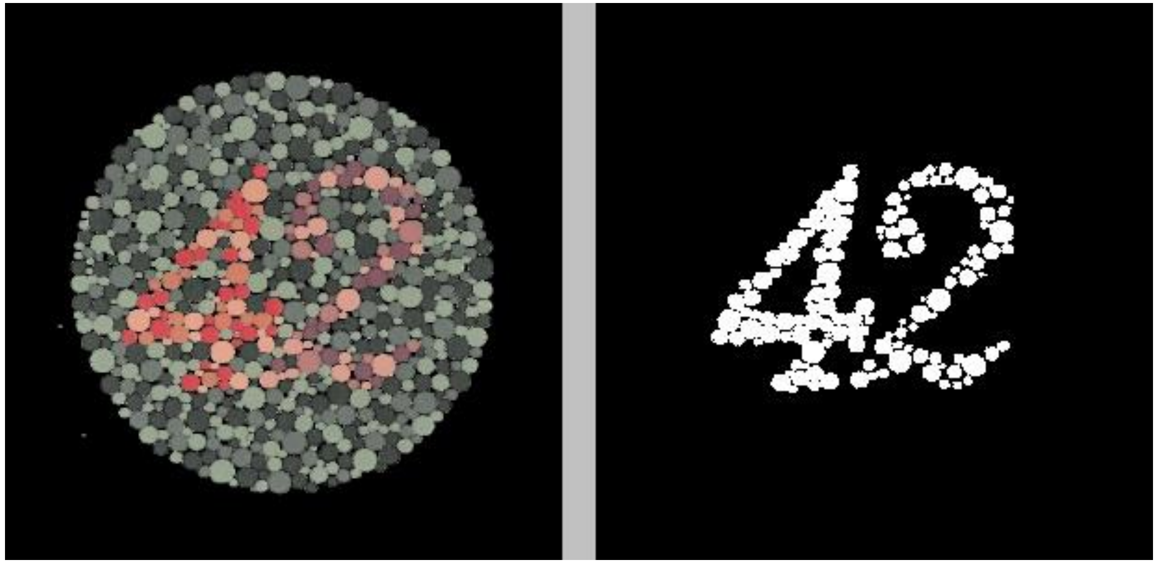


Clustering

Organizing data into clusters such that there is

- high intra-cluster similarity
- low inter-cluster similarity

Informally, finding natural groupings among objects.



Why it's needed?

- Information extraction
- Creating hierarchy of objects
- Data simplification set for further analysis
- Data compression
- Anomaly detection
- Feature generation

Clustering is Unsupervised

Clustering is an example of unsupervised learning

- We are not given examples of classes y
  - No specific task
  - No specific number of cluster
  - No specific quality criterion

Instead we have to discover classes in data

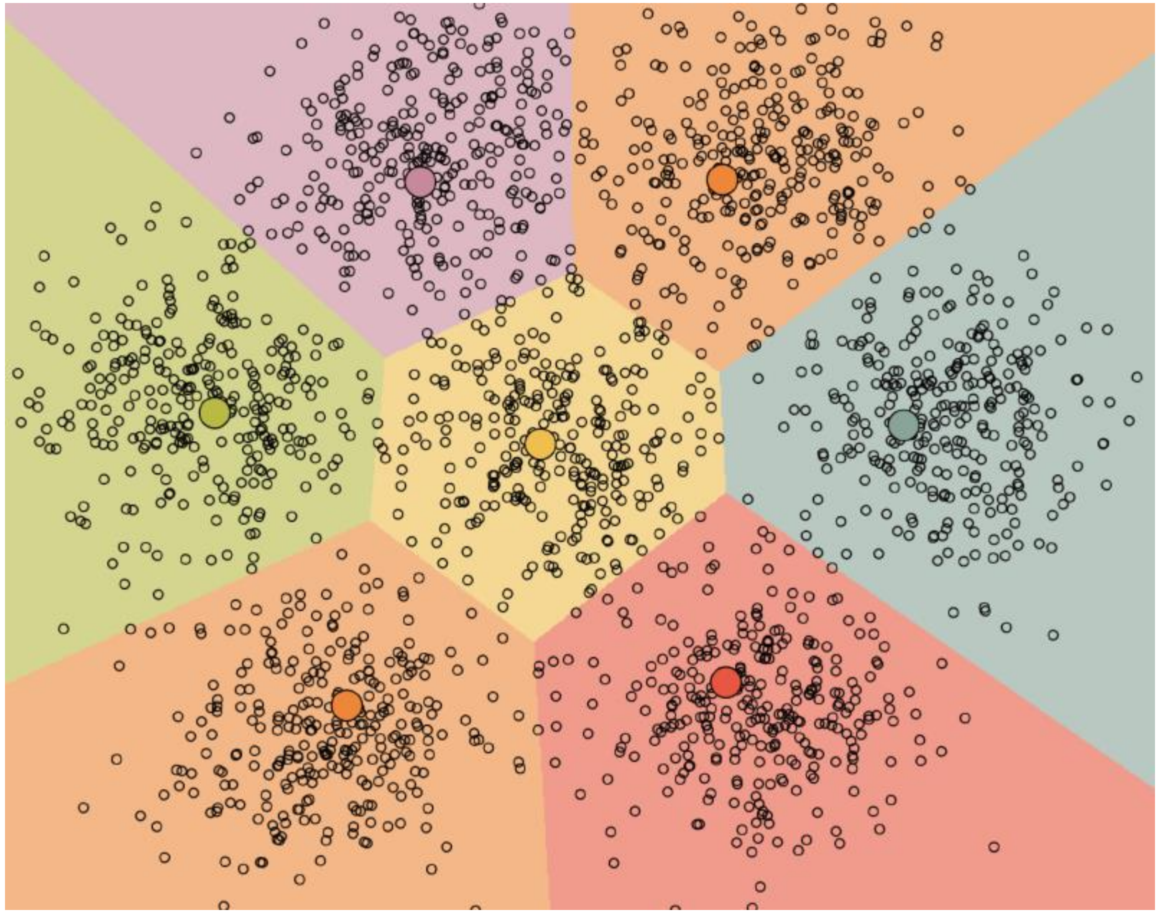
k-Means

The number of clusters is predefined

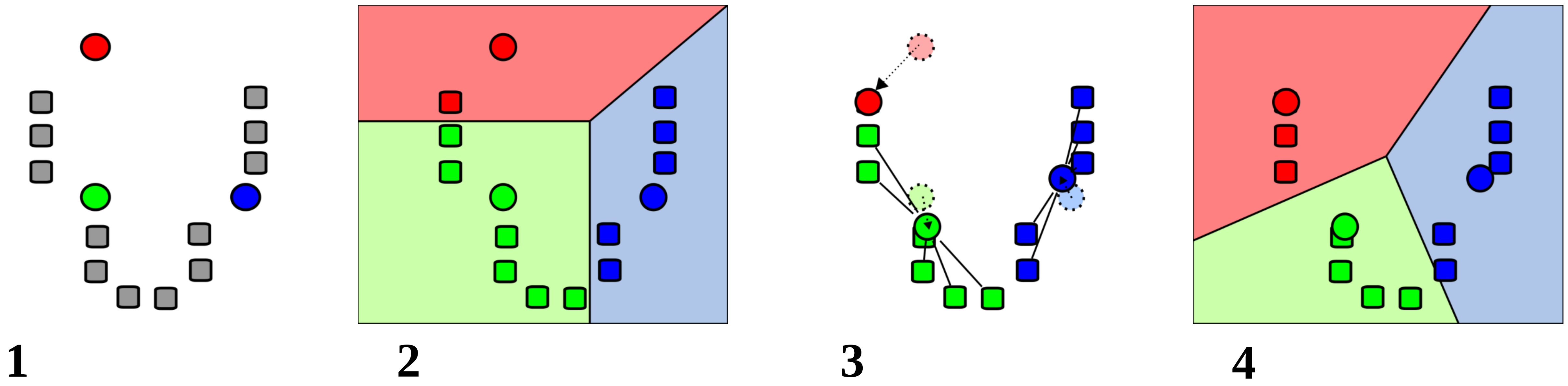
k-Means clustering aims: minimize the within-cluster sum of squares (variance)

$$\sum_{i=1}^{k} \sum_{x \in C_i} \|x - \mu_i\|_2^2 \rightarrow \min_{\boldsymbol{C}}$$

- μ_i is the center of C_i cluster
- $\boldsymbol{C}$ is the $\{C_1, C_2, \dots, C_k\}$



k-Means

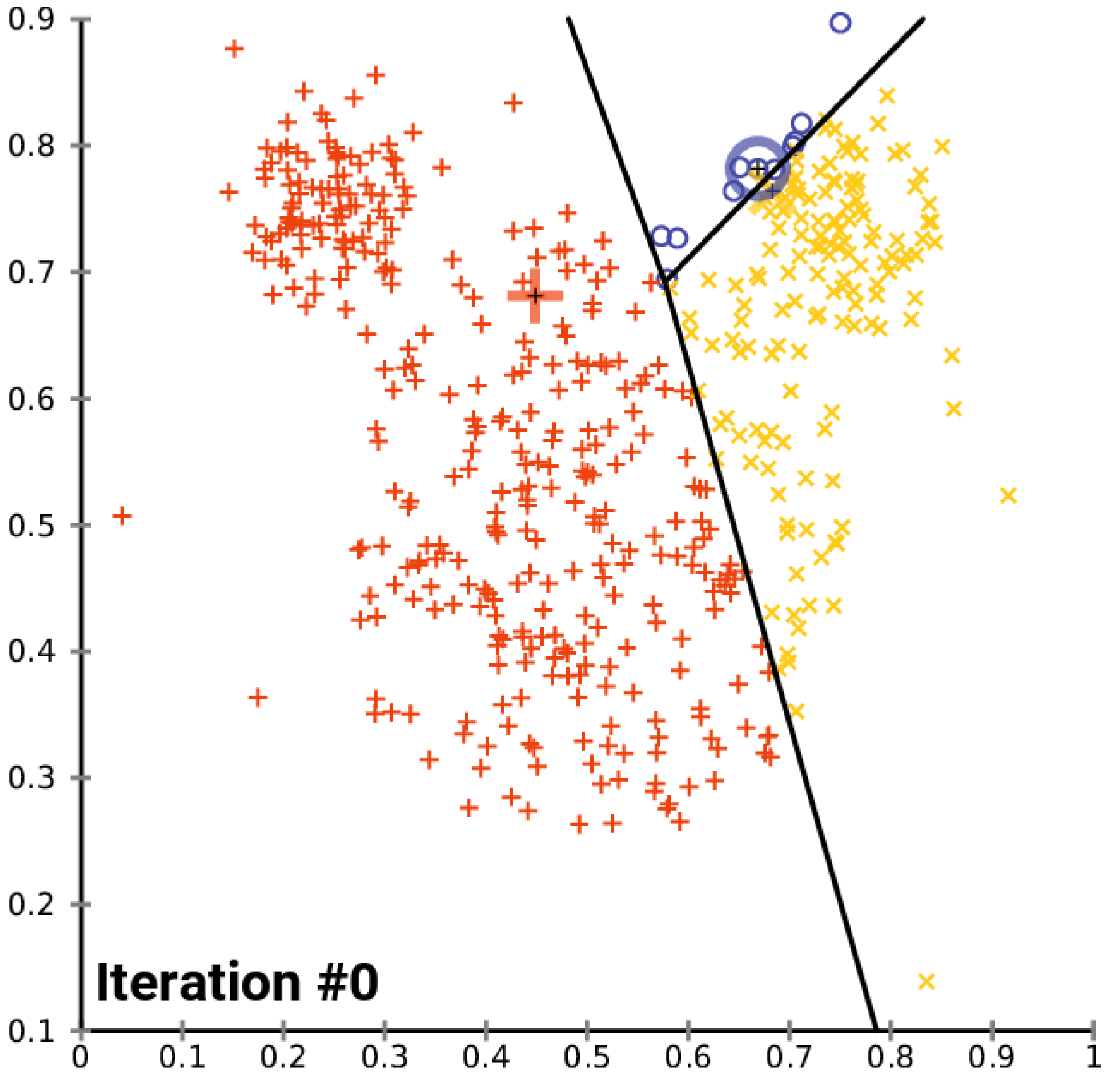


k-Means Algorithm

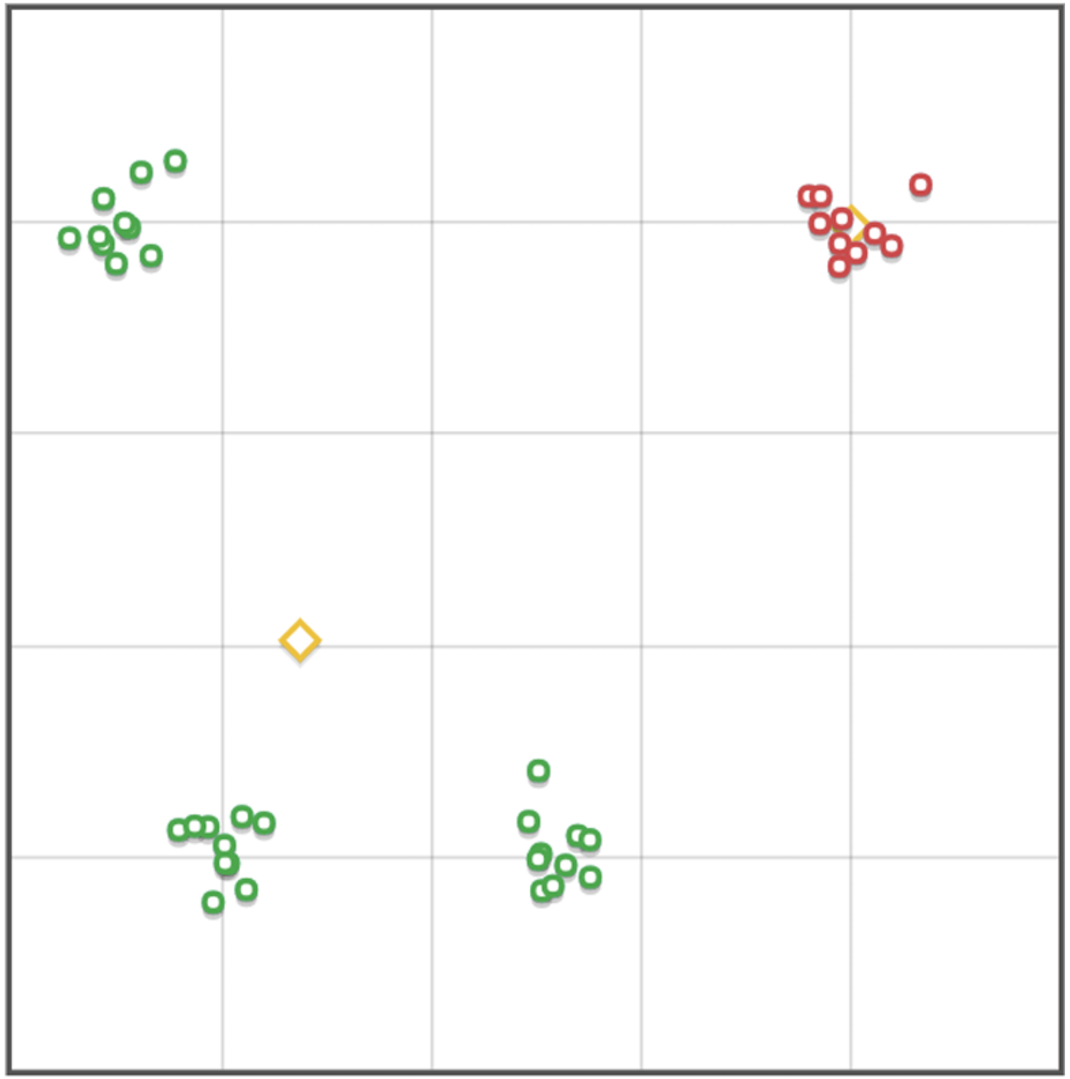
1. Initialize clusters at random.
2. Assign each point to the closest centroid
3. Shift cluster's center to the center of mass:

$$\mu_i = \frac{1}{|C_i|} \sum_{x_j \in C_i} x_j$$

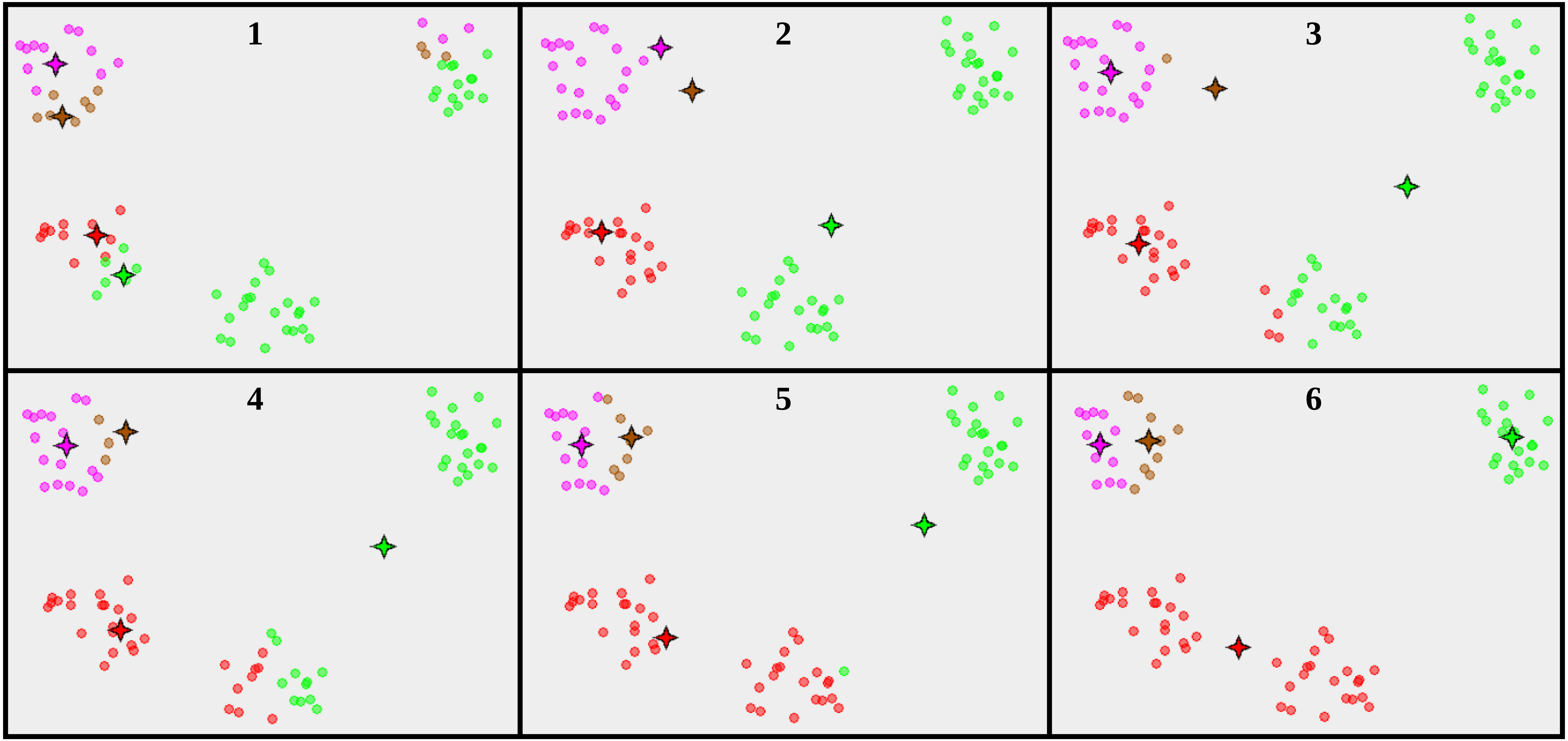
4. Repeat step 2-3 until convergence



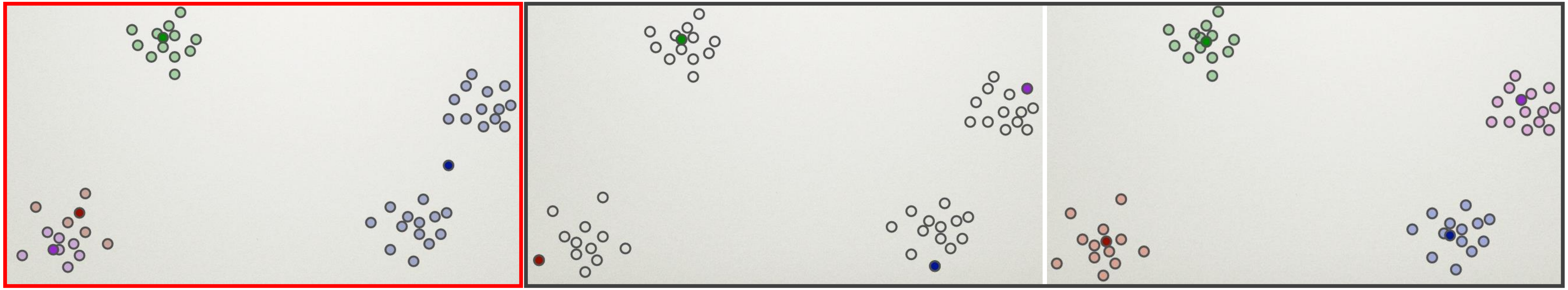
Problems: Number of Clusters

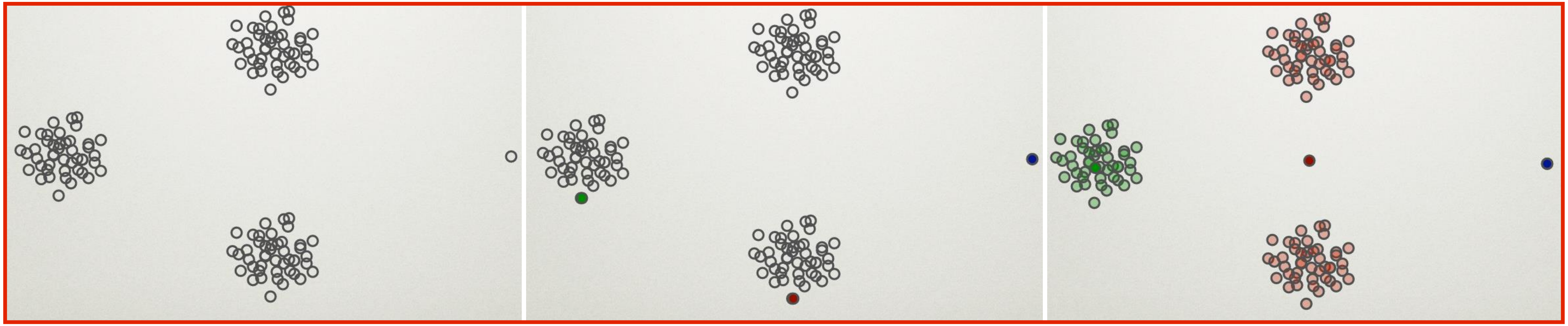


Problems: Initialization of Centers



Initialization of Centers: Easy Fix





k-Means ++

1. Choose one center uniformly at random among the data points
2. For each data point x not chosen yet, compute $D(x)$, the distance between x and the nearest center that has already been chosen
3. Choose one new data point at random as a new center, using a weighted probability distribution where a point x is chosen with probability proportional to $D(x)^2$
4. Repeat Steps 2 and 3 until k centers have been chosen
5. Launch k-means

$D(x)^\alpha$: $\alpha = 0$ – basic-like k-means, $\alpha = \infty$ – furthest point

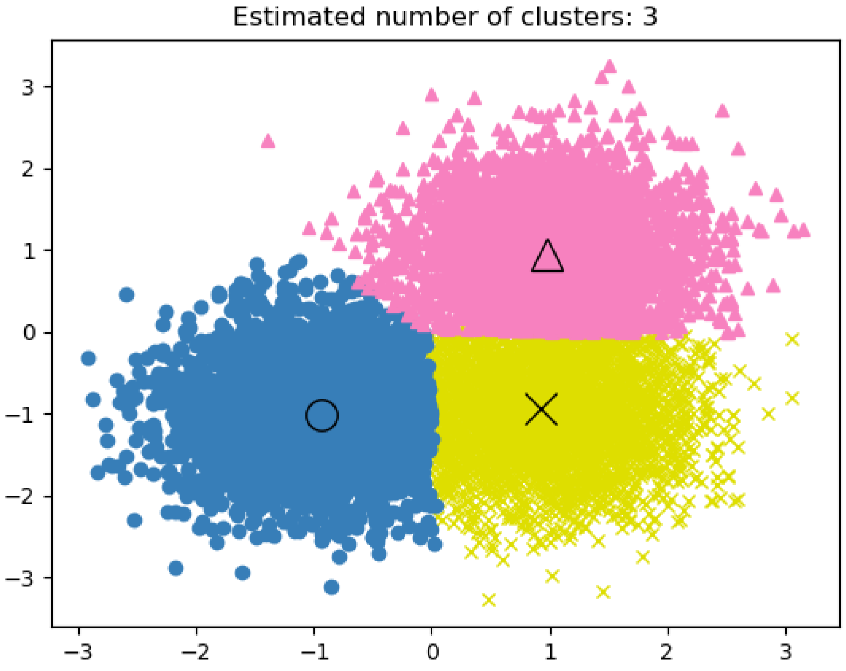
Mean Shift

The number of clusters is not predefined

1. $\mu_i^0 = x_i$

2. $\mu_i^{t+1} = \dfrac{\sum_{x_j \in N(\mu_i^t)} K(x_j - \mu_i^t) x_j}{\sum_{x_j \in N(\mu_i^t)} K(x_j - \mu_i^t)}$

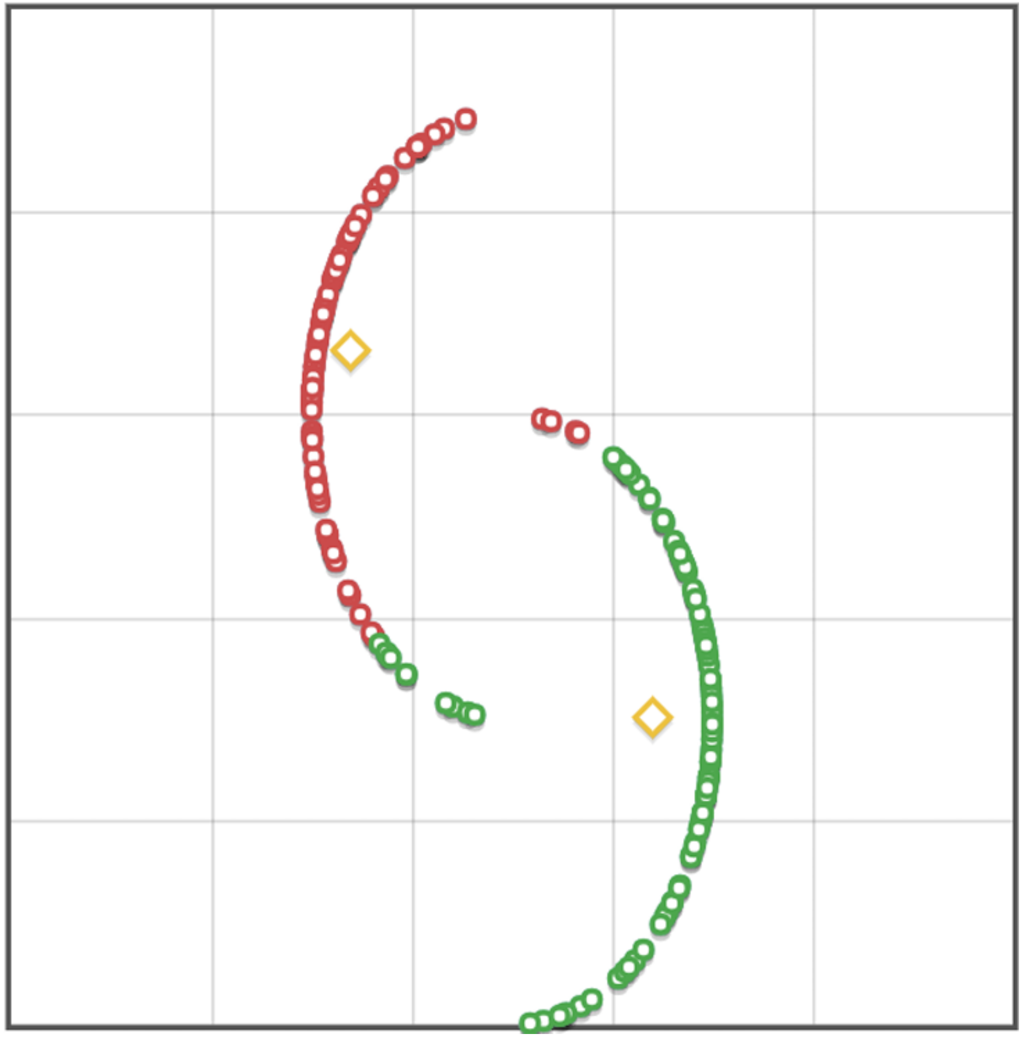
Repeat 2 until convergence, then merge close centers

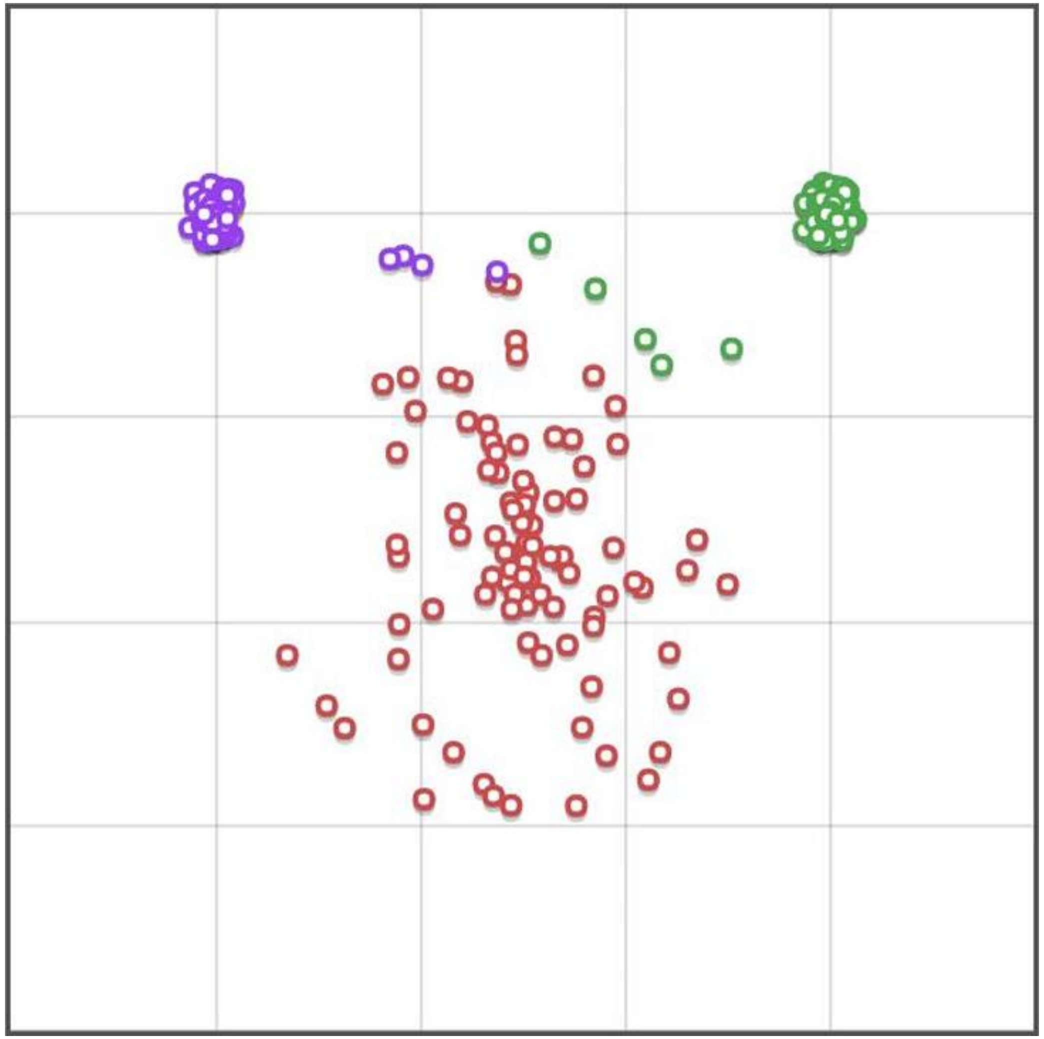


$N(\mu_i^t)$ – is some set of neigborhoods for μ_i^t. K is usually RBF – radial basis function

$$RBF(x_j - \mu_i^t) = e^{-c \left\| x_j - \mu_i^t \right\|_2^2}$$

Problems





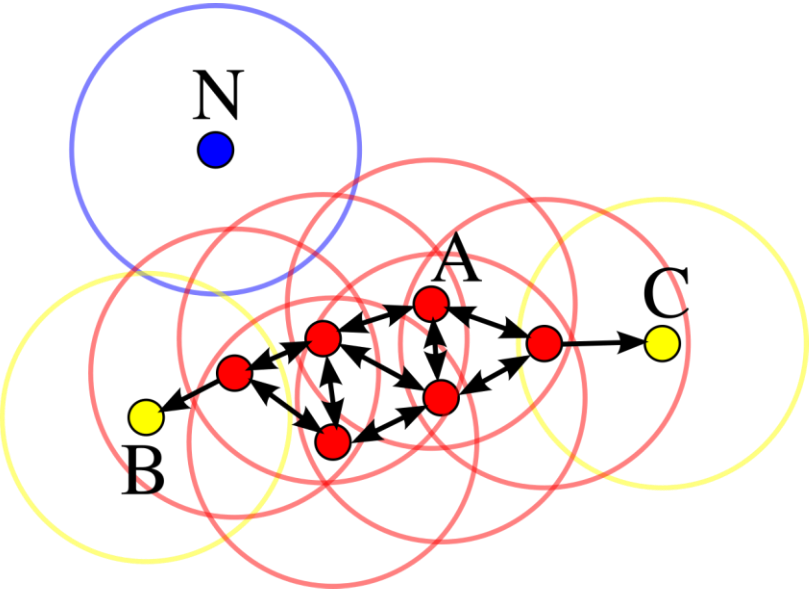
DBSCAN

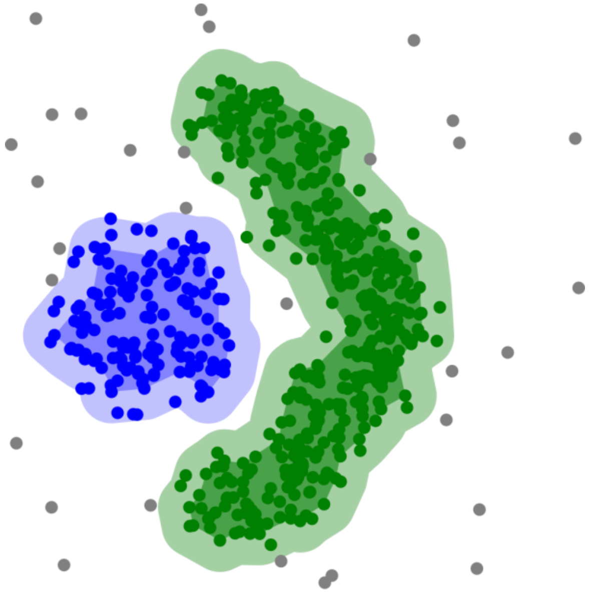
(Density-Based Spatial Clustering of Applications with Noise)

The number of clusters is not predefined.

1. Define core point – data points that contain at least m data points in their ϵ-neighborhood
2. Merge core samples (if they are within ϵ of each other) and their neighborhoods
3. Assign each non-core point to a nearby cluster if the cluster is an ϵ neighbor, otherwise assign it to noise

m = 4





Hierarchical Clustering: Agglomerative Clustering

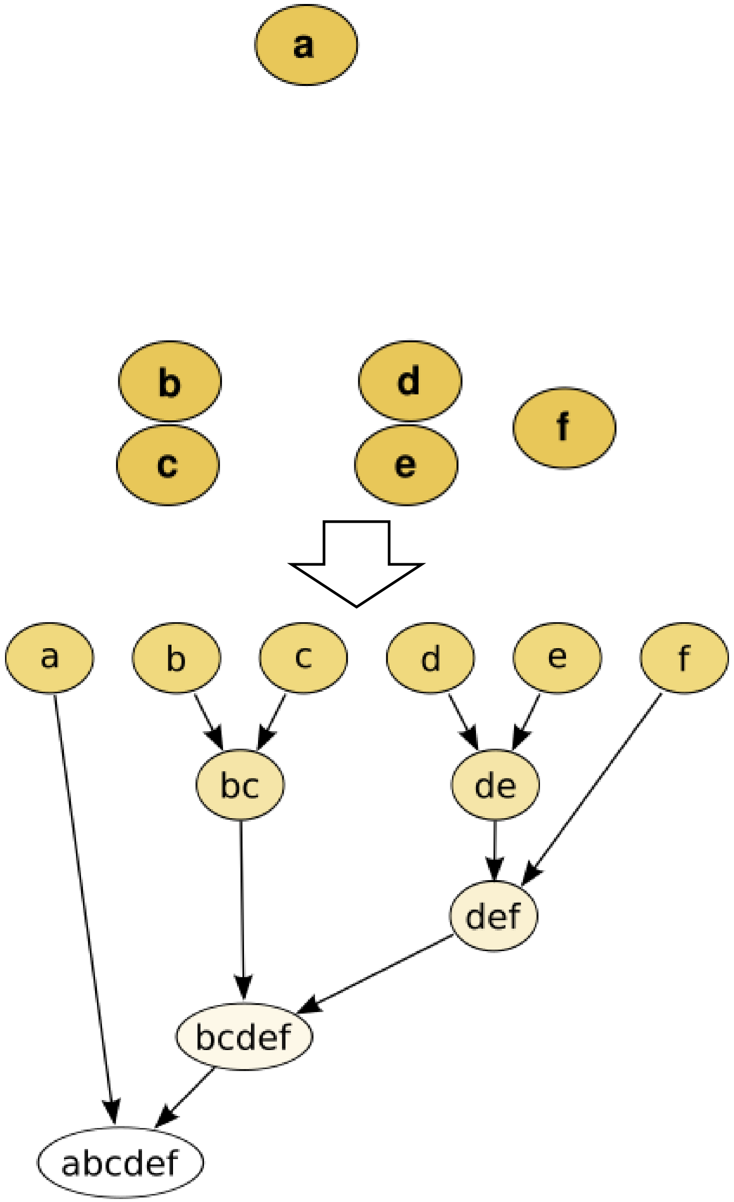
Hierarchical bottom-to-top clustering.

- Initialize every point as its own cluster.
- Three linkage strategies. Each strategy aims to minimize on of the following metrics:

  ward – variance of joined clusters

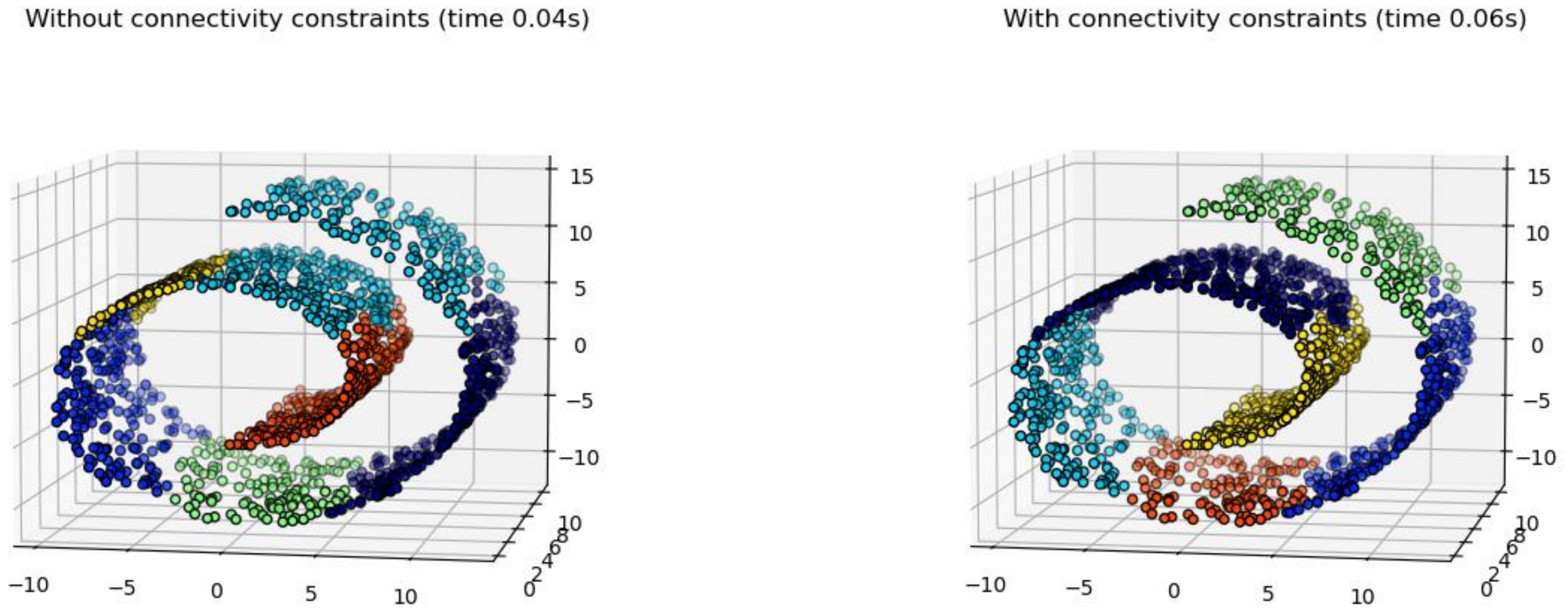
  average – average distance between points in clusters

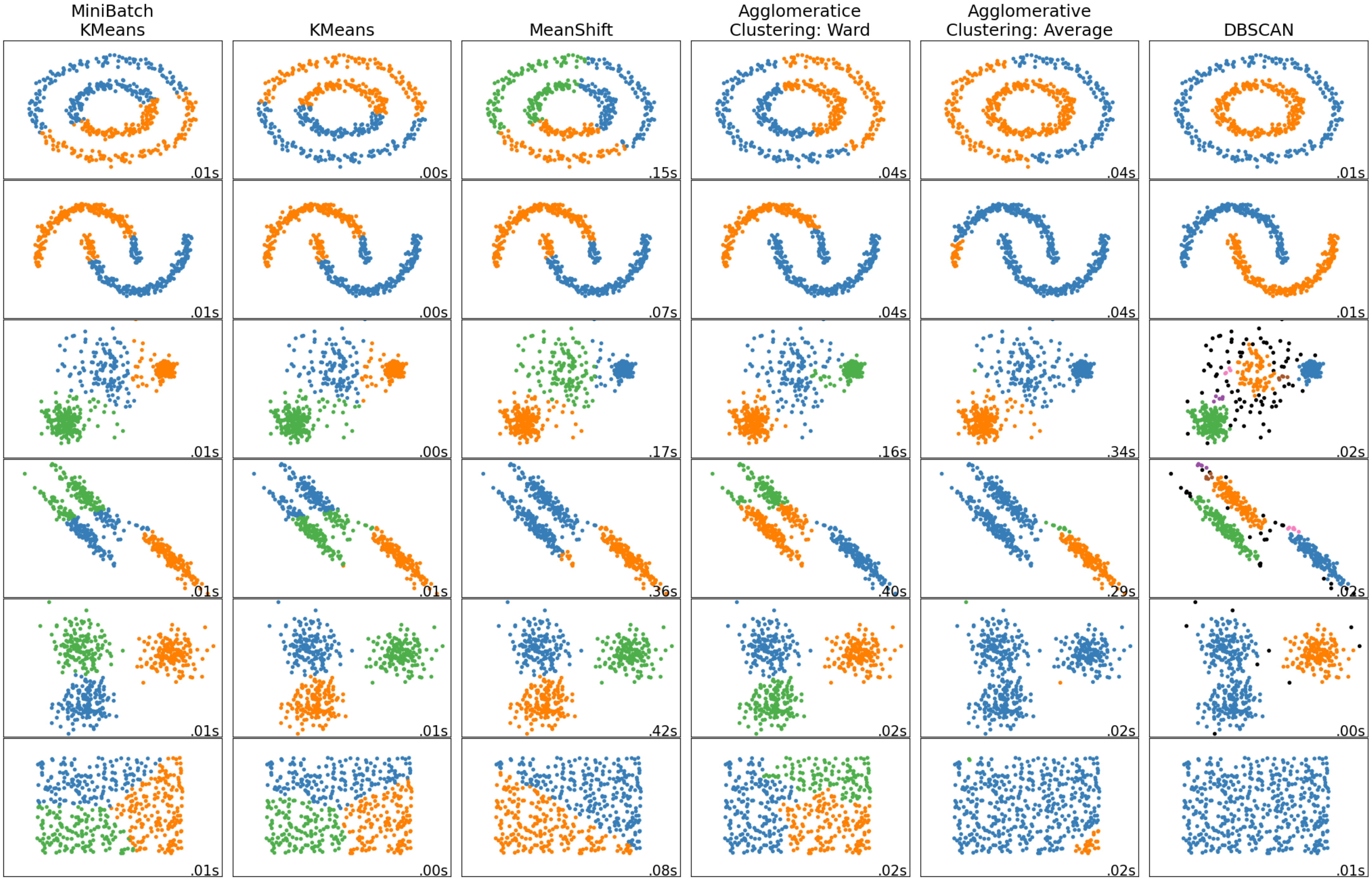
  maximum (complete) – maximum distance between points in clusters
- Join until just one cluster remains



As an additional condition you can add connectivity constraint– meaning we only join clusters if the minimum distance between them is less then some predetermined constant.

Agglomerative Clustering: Connectivity Constraint




MiniBatch KMeans
KMeans
MeanShift
Agglomeratice Clustering: Ward
Agglomerative Clustering: Average
DBSCAN
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Clustering Metrics

External:

- End metric

- Homogenity score: $\frac{1}{|D|} \sum_i \max_y |x_j \in C_i, y_i = y|$
- Rand score: $R = \frac{a+b}{a+b+c+d} = \frac{a+b}{\binom{n}{2}} = \frac{TP+TN}{TP+FP+FN+TN}$

  a – the number of pairs that lie in the same cluster in both cases

  b – the number of pairs that lie in different clusters in both cases

  c – the number of pairs that are in the same cluster, although they should be in different

  d – the number of pairs that lie in different clusters but should be in the same cluster
- Mutual information

Clustering Metrics

Internal:

- Silhouette coefficient: $s = \frac{b - a}{\max(a,b)}$

a – the mean distance between a sample and all other points in the same cluster

b – the mean distance between a sample and all other points in the next nearest cluster

- Dunn index: $D = \frac{\min_{i \neq j} \rho(\mu_i, \mu_j)}{\max_{x_i, x_j \in \mu} \rho(x_i, x_j)}$

- Davies-Bouldin index